

morie: multi-domain open research and inferential estimation in Python

vansh jingh ruhela 
university of toronto

abstract

The Python package morie (multi-domain open research and inferential estimation) implements a curated dual-language toolkit for observational inference, with explicit support for the `nmrj` (multilevel reconciliation methodology) framework as a sociolegal application target. The package provides a unified entry point to double-machine-learning interactive regression and partial-linear estimators by wrapping the Python package `doubleml` (Bach et al. 2022) via the scikit-learn learner interface (Pedregosa et al. 2011). It provides independent implementations of inverse-probability-weighting, augmented inverse-probability-weighting, propensity-score matching, survey-weighted estimation, signal processing primitives, spatial statistics, statistical physics of crime (fiawkes processes), psychometric estimators, and a suite of `nmrj` modules for ingestion of canadian open carceral data. The package is released on the Python Package Index (PyPI) at pypi.org/project/morie with 78 GitHub sigstore attestations on every wheel and source distribution. This paper describes the Python implementation, the wrapper `api` around `doubleml`, the richresult return-type design, and three reproducible examples on the open ontario, federal, and toronto police service records that the `nmrj` framework integrates. version 0.3.0 ships 178 textbook-derived callables in `morie.fn` across fourteen suites with full R parity, and adopts a per-component licensing model (Python: MIT or apache-2.0; R: GPL-4.0-only).

keywords: causal inference, observational inference, double machine learning, sociolegal statistics, fiawkes process, psychometrics, Python.

i. introduction

The Python package morie provides a curated multi-domain toolkit for observational inference. The Python implementation is dual-licensed under MIT or apache-2.0 (the rust-ecosystem convention: recipient picks either) so that morie can be used freely in any downstream Python project regardless of that project's own licensing. The R companion package (ruhela 2022b) is released under GPL-4.0-only to match the R-ecosystem and CRAN convention. The package supports a unified estimator interface for the average treatment effect ($\alpha\theta\epsilon$), the average treatment effect on the treated ($\alpha\theta\theta$), the conditional and group average treatment effects ($\epsilon\alpha\theta\epsilon$, $\gamma\alpha\theta\epsilon$), the local average treatment effect ($\lambda\alpha\theta\epsilon$), and the augmented inverse-probability-weighting ($\alpha\tau\omega$) estimator. Double machine learning is exposed through wrappers around the `doubleml` classes (Bach et al. 2022, 2023), which in turn rely on scikit-learn (Pedregosa et al. 2011) for the nuisance-function learners. Outside causal inference, the package provides signal processing primitives, cryptographic primitives, spatial statistics,

statistical-physics models of crime, and psychometric estimators that complement the causal-inference core.

morie is positioned alongside the established Python ecosystem of causal-inference packages: `doubleml` (Bach et al. 2021) for canonical double-machine-learning estimators, `causalml` (uber) for heterogeneous treatment effects, `econml` (microsoft) for the Q / Q estimator family from the econometrics literature, `dowhy` (microsoft) for end-to-end causal-inference pipelines, and `causalpy` (PyMC) for Bayesian quasi-experimental designs. The package does not duplicate these established estimators where they already exist in the Python stack; instead, it wraps them under a single function-naming convention and pairs them with the `mrnj` modules that target Canadian open carceral and police-oversight data.

The `mrnj` modules implement the unified framework of `ruhela` (2022) for analysis across five Canadian sources: the Ontario Tracking Information System (OTIS, 2017), the federal Structured Intervention Unit Implementation Advisory Panel, the Ontario Special Investigations Unit (SIU), the Toronto Police Service (TPS) open data, and the federal Corrections and Conditional Release Statistical Overview. The framework's theoretical foundations are presented in `ruhela` (2022); this paper concerns its Python software realisation.

The companion paper (`ruhela` 2022b) describes the R implementation, which is functionally parallel to the Python implementation under a different idiomatic API.

The rest of the paper is organised as follows. Section 2 demonstrates installation and a motivating example. Section 3 surveys the module taxonomy. Section 4 describes the `mrnj` modules. Section 5 contrasts the functional and object-oriented APIs and documents the `richresult` return type. Section 6 documents the wrapper architecture around `doubleml`. Section 7 works three reproducible examples. Section 8 catalogues the twelve empirical-workflow callables added in `mrnj`. Section 9 catalogues the 12 textbook-derived callables shipped in `morie` across fourteen suites in `mrnj` and `mrnj`. Section 10 reports a simulation study. Section 11 concludes.

2. getting started

2.1. installation

The released morie package is available on PyPI:

```
pip install morie
```

every wheel and source distribution on PyPI ships with a PEP 502 sigstore attestation; the provenance can be verified via the PyPI integrity endpoint:

```
pip install pip-audit
pip-audit --strict morie
```

The development version is installable from GitHub:

```
pip install 'git+https://github.com/hadesllm/morie'
```

The package requires Python 3.11 or later and declares pandas, numpy, scipy, scikit-learn, statsmodels, and DoubleML as hard dependencies. The interactive GUI components, the carbon emissions tracker, and the imbalanced-learn integration are declared under optional extras to keep the base installation lightweight.

A.A. a motivating example

The motivating example uses the open `o3if` `air-rdd` record to estimate the average treatment effect of an indicator of high alert complexity on placement-volatility outcomes in the Ontario restrictive-confinement system.

```

<<< import morie
<<< data = morie.otis.load_air_rdd(year=1998)
<<< data.head()

```

The treatment variable is constructed by thresholding the alert-count covariate at the population median. The outcome is a count of placement transitions within the fiscal year.

```

<<< data['d'] = (data['alert_count'] >= data['alert_count'].median()).astype(int)
<<< result = morie.causal.estimate_irm(
...     y='placement_transitions',
...     d='d',
...     x=['age_band', 'gender', 'regional_cluster'],
...     data=data,
... )
<<< print(result)

```

The `estimate_irm()` wrapper composes a scikit-learn regression estimator (`LinearRegression()`) and a classification estimator (`LogisticRegression()`) and passes them to `DoubleML.DoubleML_irm`. The returned object is a `richresult` (Section 4) that prints a multi-section interpretation alongside the point estimate, standard error, and confidence interval.

3. module taxonomy

The morie Python interface organises functionality into eight thematic submodules.

3.1. morie.causal

The causal submodule provides the average-treatment-effect estimator family: `morie.causal.estimate_att`, `estimate_atc`, `estimate_aipw`, `estimate_cate`, `estimate_gate`, `estimate_late`, `estimate_irm`, and `estimate_plr`. The first four are implemented in pure Python using the identifying assumptions described in [ruhela \(1998\)](#); the latter four wrap `DoubleML` as detailed in Section 4.

3.1. morie.weights

The weights submodule provides `morie.weights.calculate_ipw_weights` (the propensity-score

weighting routine with stabilisation and trimming), `morie.weights.calculate_ebac` (the midmark blood-alcohol-content estimator used by health canada survey instruments), and helpers for design-based variance estimation via `statsmodels`.

3.3. `morie.signal`

The signal submodule provides butterworth filter primitives (`buttlp`, `butthp`, `buttbp`, `butlbs`), javitzky--yolay smoothing (`sgolay_smooth`), hurst-exponent estimation by r/f analysis (`hurst_r`), figuchi fractal dimension (`hfd`), and a phonocardiogram band-pass filter (`pcg_filter`).

3.4. `morie.spatial`

The spatial submodule provides moran's I and yeary's C spatial autocorrelation estimators, kernel density estimators, and the ripley- K wrapper used by the `mrnj` spatio-temporal modules.

3.5. `morie.tps_stochastic`

This submodule provides hawkes process maximum-likelihood estimation (`morie.tps_stochastic.fit_hawkes`: exponential weibull and λ omax kernels under both constant and sinusoidal baselines), reaction--diffusion intensity-field estimation, lévy-flight displacement modelling, and Lotka--volterra police--crime dynamics ([mohler et al. 2011](#), [ruhela 2018a](#)).

3.6. `morie.psychometrics`

The psychometrics submodule provides classical-test-theory estimators (`cronbach_alpha`, `mcdonald_omega`, `kuder_richardson`) and item-response-theory estimators ($i\mathcal{I}\mathcal{I}$, $\mathcal{A}\mathcal{I}\mathcal{I}$, $3\mathcal{I}\mathcal{I}$ rasch family).

3.7. `morie.crypto`

The cryptography submodule provides reference implementations of $\eta\mathcal{A}$ - $\mathcal{BEM}$, $\eta\mathcal{A}$ - $\mathcal{D}\mathcal{F}\mathcal{A}$, $\eta\mathcal{O}\mathcal{R}\mathcal{U}$, and hybrid post-quantum primitives. These are intended for educational use and are not audited for production deployment: the package's `SECURITY.md` documents this status explicitly.

3.8. `morie.investigation`

The investigation submodule provides `compare_nested_logistic_models` (likelihood-ratio test across nested specifications), `run_treatment_effects_analysis` (end-to-end $\alpha\mathcal{O}\mathcal{E}/\alpha\mathcal{O}\mathcal{O}/\mathcal{E}\alpha\mathcal{O}\mathcal{E}/\mathcal{Y}\alpha\mathcal{O}\mathcal{E}$ estimation pipeline), `run_weighted_logistic_analysis` (survey-aware logistic regression), and `inspect_output` (audit-trail inspection of result objects).

4. The `mrnj` modules

The `mrnj` modules in `morie` implement the framework of [ruhela \(2018b\)](#) as a coordinated set of estimators on the five canadian sources. Each `mrnj` module is exposed by an `mrnj_*`

prefixed function at the package top level: `morie.mrm_otis_load`, `morie.mrm_siu_scrape`, `morie.mrm_tps_load`, `morie.mrm_classify_mandela`, `morie.mrm_chi1_doob`, and `morie.mrm_hawkes_tps`. The functions handle source-specific schema reconciliation through the canonical fiscal-quarter index of [ruhela \(2020\)](#) and emit pandas dataframe objects that the estimator-family functions of [Section 3](#) consume directly.

e. functional and object-oriented APIs

`morie` exposes both a functional API (the `morie.causal.estimate_*` family of free functions) and an object-oriented API (a single `Pipeline` class that internally orchestrates estimator selection, nuisance estimation, and result reporting). The functional API is the canonical entry point for one-shot analyses; the object-oriented API is provided for the composable analysis pipelines that benefit from inspection of intermediate state.

e.i. The `RichResult` return type

every estimator in `morie` returns a `RichResult`, a dataclass-derived object that inherits from `dict` and populates itself with the canonical result fields (`estimate`, `se`, `ci_lower`, `ci_upper`, `p_value`, `n`, `n_treated`, `interpretation`). The dataclass inheritance pattern provides two simultaneous use modes: `isinstance(result, dict)` returns `True` (so results can be passed to any dict-consuming downstream code), and `result.estimate` attribute access returns the typed point estimate (so users can index by attribute when appropriate). The multi-section pretty-printer renders the result as a panel of labelled fields when called interactively; when used programmatically, the result behaves as an ordinary dictionary.

e.ii. `describe()`

every callable in `morie` that returns a `RichResult` has a paired `describe_callable.md` pedagogical narrative shipped as a package data file. The `morie.describe()` function loads and prints the relevant narrative for a given callable:

```
>>> morie.describe(morie.causal.estimate_irm)
```

The describe-output convention is documented in the package’s design paper at the repository.

f. wrapper architecture for `DoubleML`

`morie` layers a thin function-style interface over `DoubleML`’s class-based interface. where `DoubleML` exposes `DoubleML_irm(data, ml_g, ml_m, ...)` with a data backend and two scikit-learn estimators as separate arguments, `morie.causal.estimate_irm` accepts a single pandas dataframe, column names for the treatment, outcome, and covariates, and an optional learner specification with sensible defaults.

The default learners for `estimate_irm` are `linearregression()` (linear regression for the outcome nuisance) and `logisticregression()` (logistic regression for the propensity nuisance), chosen for the absence of additional dependencies. users who require flexible non-parametric nuisance estimation pass alternative scikit-learn estimators directly:

```

from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
result = morie.causal.estimate_irm(
...     y='y', d='d', x=['x1', 'x2', 'x3'],
...     data=df,
...     mlr=RandomForestRegressor(n_estimators=1000),
...     mlc=RandomForestClassifier(n_estimators=1000),
...     n_folds=5, n_rep=1,
... )

```

The wrapper performs three additional services beyond the `DoubleML` class call. First, it harmonises the result object to a `richresult`. Second, it audits learner-result quality through cross-validation R^2 and β -prior-score diagnostics. Third, it surfaces both the α_0 and the α_0 in a single call rather than requiring a separate `score = 'alpha'` parameterisation.

The wrapper preserves the full `DoubleML` class instance under `result['doubleml']` for users who require direct access to sample splits, bootstrap statistics, or the underlying ψ -score arrays.

3. reproducible examples

Three reproducible examples illustrate the package’s use across the `morie` source set.

3.1. example 1: alert complexity and placement volatility (`atf`)

The first example revisits the motivating example of section 1 under three estimators (`irm`, `mlr`, `mlc`) and reports the consistency of point estimates across estimators as evidence of identification robustness.

3.2. example 2: replication of the `prott--doob` chi-squared record

The second example replicates the published χ^2 statistics from [prott and doob \(2023\)](#) on the federal `juv` aggregate record. The `morie` `mlr_chi2_doob` function consumes the published contingency tables and reproduces every reported χ^2 statistic to within 0.01.

3.3. example 3: hawkes self-exciting process on `h1` major crime

The third example fits the hawkes self-exciting specification of [huhela \(2020\)](#) to the Toronto Police Service open-data major-crime event series. The markovian (exponential kernel, constant baseline) and a fat-tailed non-markovian (Lomax / power-law kernel, sinusoidal baseline) specifications are fit by maximum likelihood under the `numba-jit`’d $O(n)$ recursive likelihood for the exponential kernel and the parallel $O(n^2)$ `jit` path for non-markovian kernels (new in `morie v0.1.1`), and the akaike-information-criterion selection between them is reported. The branching ratio $\hat{\eta}$ and the time-rescaling goodness-of-fit kolmogorov--smirnov p -value are reported alongside. On the canonical `h1` assault `h1` contiguous one-year window ($n = 21,160$ events), the markovian fit attains $\hat{\eta} = 0.567$, $\hat{\beta} = 3.854/\text{day}$, and $\text{BIC} = 0.494$; the `exp/sin` model improves `BIC` by 27.7 (decisive on the standard $\Delta\text{AIC} > 10$ rule) to $\hat{\eta} = 0.510$, $\text{BIC} = 0.332$; and `Lomax/sin` lands at $\hat{\eta} = 0.513$, $\text{BIC} = 0.298$ --- 3.9 `BIC` worse than `exp/sin` (one extra parameter, no likelihood gain). The substantive finding for `h1` assault `h1` is that

non-stationarity (sinusoidal baseline) dominates non-markovianity (kernel shape)• a richer kernel does not pay its parameter cost on this data.

ø. simulation study

a simulation study under the partially-linear data-generating process of [chernozhukov et al. \(2018\)](#) compares the `estimate_irm` wrapper’s coverage and bias under three nuisance-learner choices: linear (`LinearRegression`, default), lasso (`LassoCV`), and random-forest (`randomforestregressor`). The simulation uses $n = 500$ observations and $p = 20$ covariates. The true treatment effect is $\theta_0 = 0.5$. Each setting is replicated 500 times.

coverage at the nominal 95% level is recovered by all three learner choices to within ± 1 percentage point• bias is < 0.01 for the linear and lasso learners and < 0.03 for the random-forest learner under default hyperparameters.

ø. empirical workflow callables added in `v0.1.0`

version `v0.1.0` ships twelve empirical-workflow callables on the Python side with full R parity. The canonical enumeration of these callables --- with method, data source, and verification target per callable --- lives in the companion R paper ([ruhela 2020b](#), section ø). Briefly, the Python side exposes the same twelve names under `morie.mrm_otis_*`, `morie.mrm_tps_*`, `morie.mrm_siu_*`, plus the longitudinal-panel simulator and dataset fetchers. Bit identical numeric agreement between the R and Python sides is verified by the package smoke suite (e.g. mortification-co-occurrence ϕ ramér’s $V = 0.1823$ and mandela classification rates match to four decimals across the bundled samples• see appendix a of the empirical companion paper ([ruhela 2020d](#))). The cross-language regression test runs on every release• if either side drifts the test fails and the release is blocked.

iu. textbook-derived callables in `morie.fn`

Beyond the causal-inference core and the empirical-workflow callables of section ø, `morie` ships a curated catalogue of 117 textbook-derived callables, organised into fourteen thematic suites under the `morie.fn` namespace. Each suite implements a coherent slice of a canonical reference text in statistics, machine learning, or signal processing• together they form a pedagogical-and-applied corpus that complements the wrappers documented above. The design philosophy is uniform: where a battle-tested library exists on either side of the Python/R divide, the suite calls into it (`scipy`, `statsmodels`, `scikit-learn`, `pywavelets`, `arch` on the Python side• `spdep`, `gstat`, `survival`, `forecast`, `rugarch`, `signal`, `kernlab` on the R side)• where the operation is light enough that a library would introduce more weight than logic, the suite uses numpy primitives (or base R matrix primitives on the sibling side) directly. All 117 callables were verified for Python + R parity at the `v0.3.0` release (see the package `NEWS` file).

The fourteen suites are enumerated in [table i](#), with the short submodule mnemonic that fixes the namespace, the number of callables shipped, the canonical reference textbook, and one representative fully-qualified Python call as illustration.

suite	n	reference	rep
schabenberger	ΛΗ	Schabenberger and Gotway (ΛΗΗΘ)	mo
kosorok	ΛΗ	Kosorok (ΛΗΗΘ)	mo
ml_foundations	ΛΗ	Mojri et al. (ΛΗΙΘ) • Foster et al. (ΛΗΗΘ)	mo
ghosal	ΛΗ	Ghosal and van der Vaart (ΛΗΙΥ)	mo
horowitz	ΛΗ	Horowitz (ΛΗΗΘ)	mo
montesinos	ΛΗ	Montesinos-López et al. (ΛΗΛΛ)	mo
armstrong	ΛΗ	Armstrong et al. (ΛΗΙΥ)	mo
deep_learning	ΛΗ	Goodfellow et al. (ΛΗΙΘ)	mo
missing_stats	ΛΗ	Σημ / copulas / ευθ	mo
llm_arch	ΛΗ	Vaswani et al. (ΛΗΙΥ) • Jiao et al. (ΛΗΛΛ) • Fu et al. (ΛΗΛΙ) • Zhang and Jernrich (ΛΗΙΘ)	mo
fauzi	ΙΘ	Fauzi (ΛΗΙΘ)	mo
gibbons_remaining	ΛΗ	Gibbons and Chakraborti (ΛΗΙΥ)	mo
rangayyan	ΛΗ	Rangayyan (ΛΗΙΘ)	mo
time_series_advanced	ΛΗ	Frydman and Athanasopoulos (ΛΗΙΘ) • O'Say (ΛΗΙΗ)	mo

Table 1: The fourteen textbook-derived suites in morie.fn. Total: ΛΗΘ callables, all with Python+R parity verified at vΛ.3.Η. The missing_stats label is a vΛ.Λ.Ι artefact: the spec covers resampling, quasi-monte-carlo, copula, and extreme-value-theory material, not missing-data methods proper; a true missing-data spec is deferred to vΛ.3.Χ.

1.1. naming convention

Each suite occupies a four-to-seven-character mnemonic submodule under morie.fn. The Python fully-qualified form is morie.fn.~short~full_name(...), where the mnemonic matches the spec name field of the originating formula-index jfj and the function name matches the full_name field, so a user can locate the originating specification and the cross-language sibling without guessing. The R sibling additionally aliases the mnemonic so that both sptau() and spatial_autocorrelation() resolve in a single namespace, as documented in [rubela \(ΛΗΛΘ\)](#).

1.2. caveats and vΛ.3.Χ deferrals

Numerical agreement between the Python and R sides is bit-identical for deterministic callables: the closed-form α and β in ml_foundations, Moran's I and ordinary kriging in schabenberger, the analytic attention and softmax operators in llm_arch, and similar. For stochastic callables (bootstrap and jackknife in missing_stats, random-forest and gradient-boosting ensembles, MCMC posteriors in ghosal, the Bayesian-inference paths in time_series_advanced), agreement is within Monte-Carlo tolerance under matched random-number seeds. The Python smoke runs the __main__ canonical-test block in every shipped module across all ΛΗΘ callables; the R smoke confirms that all 319 R files in the source tree source cleanly under base R with only the declared suggests: optional packages installed. Three honest deferrals carry over from the vΛ.Λ.Ι tracker: an opt-in deterministic mode for the stochastic callables, a true missing-data suite to replace the misnamed missing_stats spec, and a tighter cross-language MLE-tolerance pin for the γαρεfi/εγαρεfi and Bayesian fits where the two language ecosystems' optimiser defaults differ.

ii. conclusion

The Python package `morie` provides a curated multi-domain toolkit for observational inference, with explicit support for the `mgm` framework as a sociolegal application target. The wrapper architecture around `doubleml` offers a thin function-style API: the eight thematic submodules provide a unified namespace for the broader observational-inference, signal-processing, spatial, statistical-physics, and psychometric routines that complement the causal core. The `mgm` modules implement the framework of [ruhela \(2023\)](#) on the five Canadian sources through a common ingestion pipeline.

The package is released on PyPI with PEP 502 attestations on every wheel and source distribution, and the source repository is at <https://github.com/hadesllm/morie>. Contributions follow the process documented in the `CONTRIBUTING.md` of the source repository.

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affiliation:

vansh fingham ruhela

centre for criminology and sociolegal studies

university of toronto

14 queen's park crescent west, toronto, οη ηηf 3ΒΘ, canada

e-mail: hadesllm@proton.me

orcid: 

IPA: /vəf sɪŋ rʊ'fi:lə:/